

Ex no. 1
22/7/25

Load the Titanic dataset & convert it into a Dataframe

AIM:

To perform basic preprocessing & EDA on Titanic dataset using pandas, seaborn & sklearn tools.

PROCEDURE:

1. Load the Titanic dataset into a Dataframe
2. Preview and check dataset
3. Handle missing values
4. Remove duplicates
5. Encode categorical columns
6. Standardize numerical data.
7. Perform visualizations.

CODE:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler

df = sns.load_dataset("titanic")

print("First 5 rows:\n", df.head())

print("\nMissing values:\n", df.isnull().sum())

df['age'].fillna(df['age'].median(), inplace=True)
df['cabin'].fillna("unknown", inplace=True)
```

```
df.drop_duplicates(inplace=True)
```

```
le = LabelEncoder()
```

```
df['sex'] = le.fit_transform(df['sex'].astype(str))
```

```
scaler = StandardScaler()
```

```
df[['age', 'fare']] = scaler.fit_transform(df[['age', 'fare']])
```

```
sns.pairplot(df[['pclass', 'sex', 'age', 'sibsp']])
```

```
plt.show()
```

```
sns.heatmap(df[['pclass', 'age', 'sibsp', 'parich', 'fare']].corr(),
```

```
annot=True, cmap="coolwarm")
```

```
plt.show()
```

RESULT:

* Thus the basic preprocessing & EDA on titanic dataset has been implemented successfully.

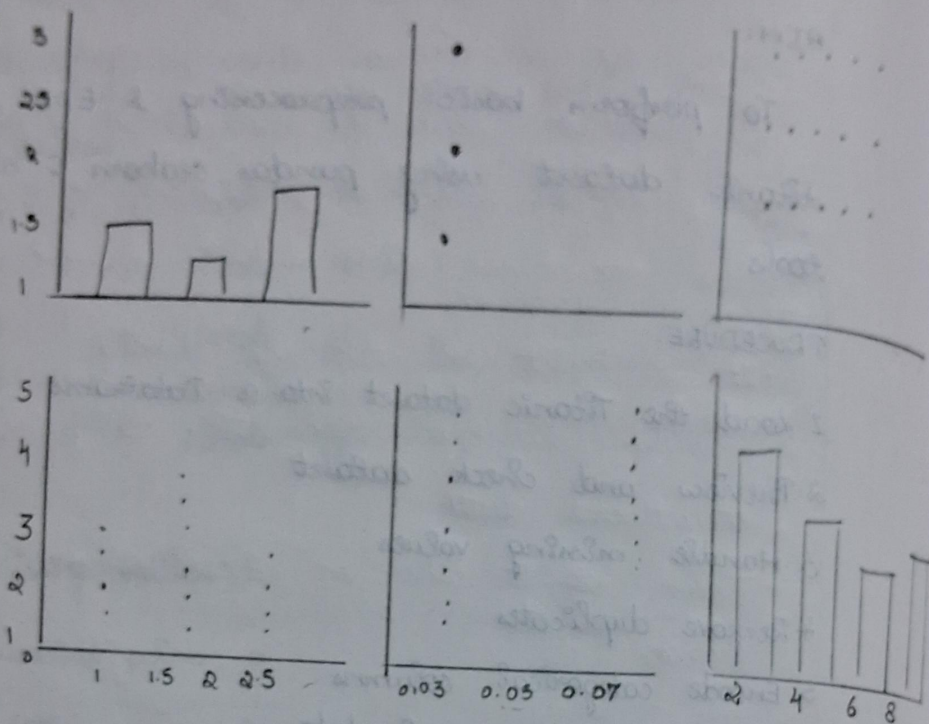
OUTPUT:

survived	sex	age	sibsp	parch	fare	class
pclass						
0 0 3	male	22	1	0	7.25	third
1 1 1	female	38	1	0	7.23	First
2 1 3	female	26	0	0	7.92	Third
3 1 1	female	35	1	0	33.1	First
4 0 3	female	35	0	0	8.05	Third

Missing values

survived	pclass	0
sex		0
age		177
sibsp		0
parch		0
embarked		2
who		0
alive		0

Pair plot:



pclass	1.00	-0.37	0.09	0.02	-0.35
age	-0.37	1.00	-0.32	-0.2	0.09
sibsp	0.09	-0.32	1.00	0.41	0.13
parch	0.02	-0.2	0.41	1.00	0.21
fare	-0.35	0.09	0.13	0.21	1.00

pclass age sibsp parch fare